

Project Design Phase
Solution Architecture

Date	1 July 2026
Team ID	
Project Name	
Maximum Marks	4 Marks

Solution Architecture – HouseRent (MERN Stack House Rental Platform)

1. Solution Overview

The HouseRent platform is a MERN Stack web application that connects House Seekers (Users), Property Owners, and Administrators. It provides secure authentication, verified property listings, property booking, image uploads, and an admin approval system. The solution uses a client-server architecture where the React frontend communicates with the Node.js/Express backend through REST APIs, and MongoDB stores the application data.

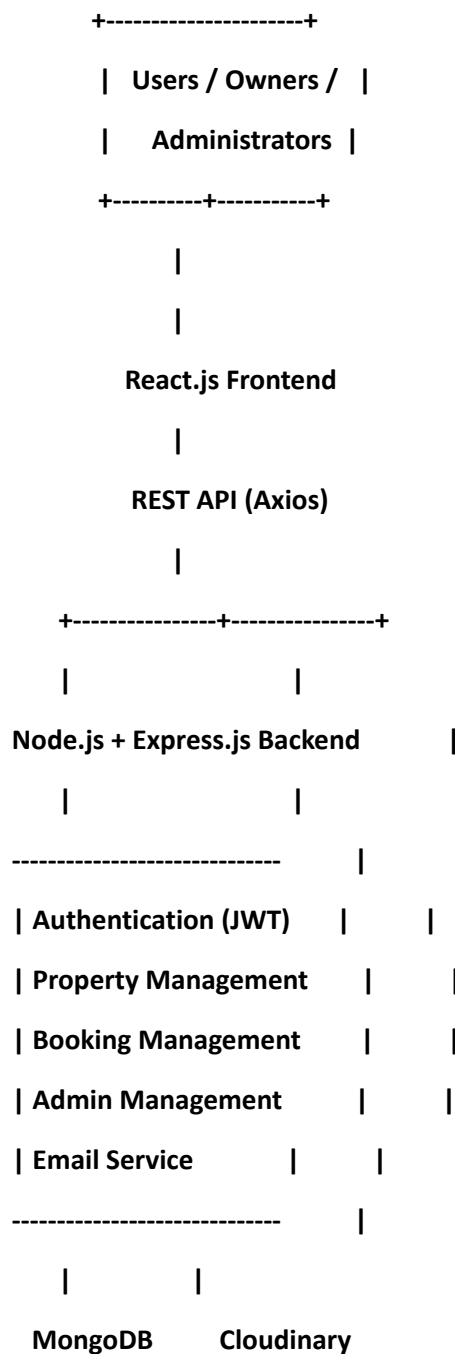
2. Architecture Components

Component	Description
Frontend (React.js)	User interface for Users, Owners, and Admin with responsive pages.
Backend (Node.js + Express.js)	Handles business logic, REST APIs, authentication, and booking management.
Database (MongoDB)	Stores users, properties, bookings, and related information.
Authentication (JWT & Bcrypt)	Provides secure login, password encryption, and role-based access.
Cloudinary + Multer	Uploads and stores multiple property images securely.
Nodemailer	Sends password reset emails to users.

3. Solution Workflow

1. User or Owner registers and logs in.
 2. Owner submits registration for admin approval.
 3. Admin approves or rejects the owner request.
 4. Approved owners add, edit, or delete property listings.
 5. Users browse available properties and view details.
 6. Users book rental properties.
 7. Booking information is stored in MongoDB.
 8. Owners view bookings for their properties.
 9. Admin manages users, owners, properties, and bookings.
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4. System Architecture



(Users, Properties, (Images)

Bookings)

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Nodemailer

(Password Reset Email)

5. Key Features

- Secure JWT Authentication
 - Role-Based Access Control (User, Owner, Admin)
 - Admin Approval for Owners
 - Property Management
 - Property Booking
 - Booking History
 - Cloudinary Image Upload
 - Forgot & Reset Password
 - Admin Dashboard
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6. Benefits of the Architecture

- Secure and scalable system.
- Modular frontend and backend.
- Easy maintenance and future enhancements.
- Fast communication through REST APIs.
- Reliable image storage using Cloudinary.
- Centralized database management with MongoDB.
- Supports future features like payments, reviews, notifications, and analytics.

Example - Solution Architecture Diagram:

